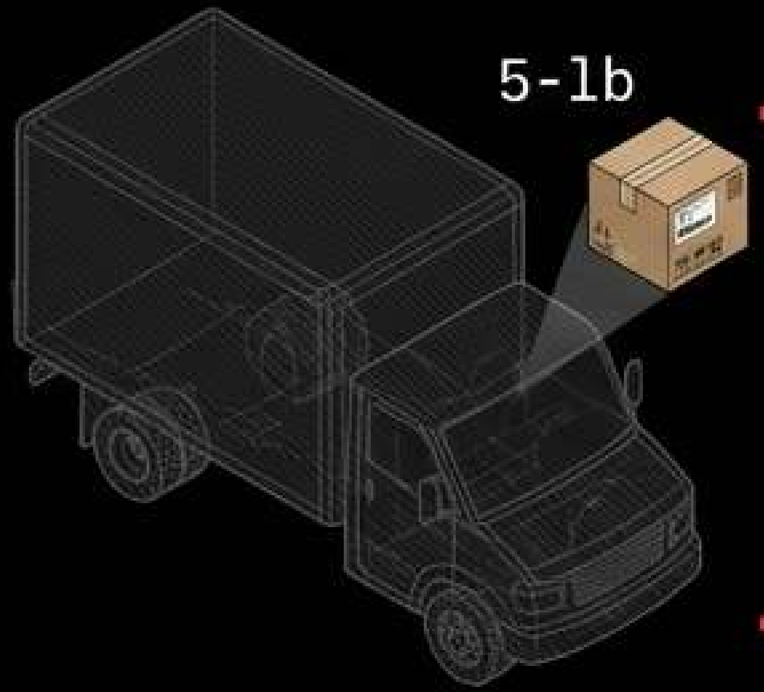
Bypassing the Physics Inefficiency of the Last Mile

The Broken Paradigm

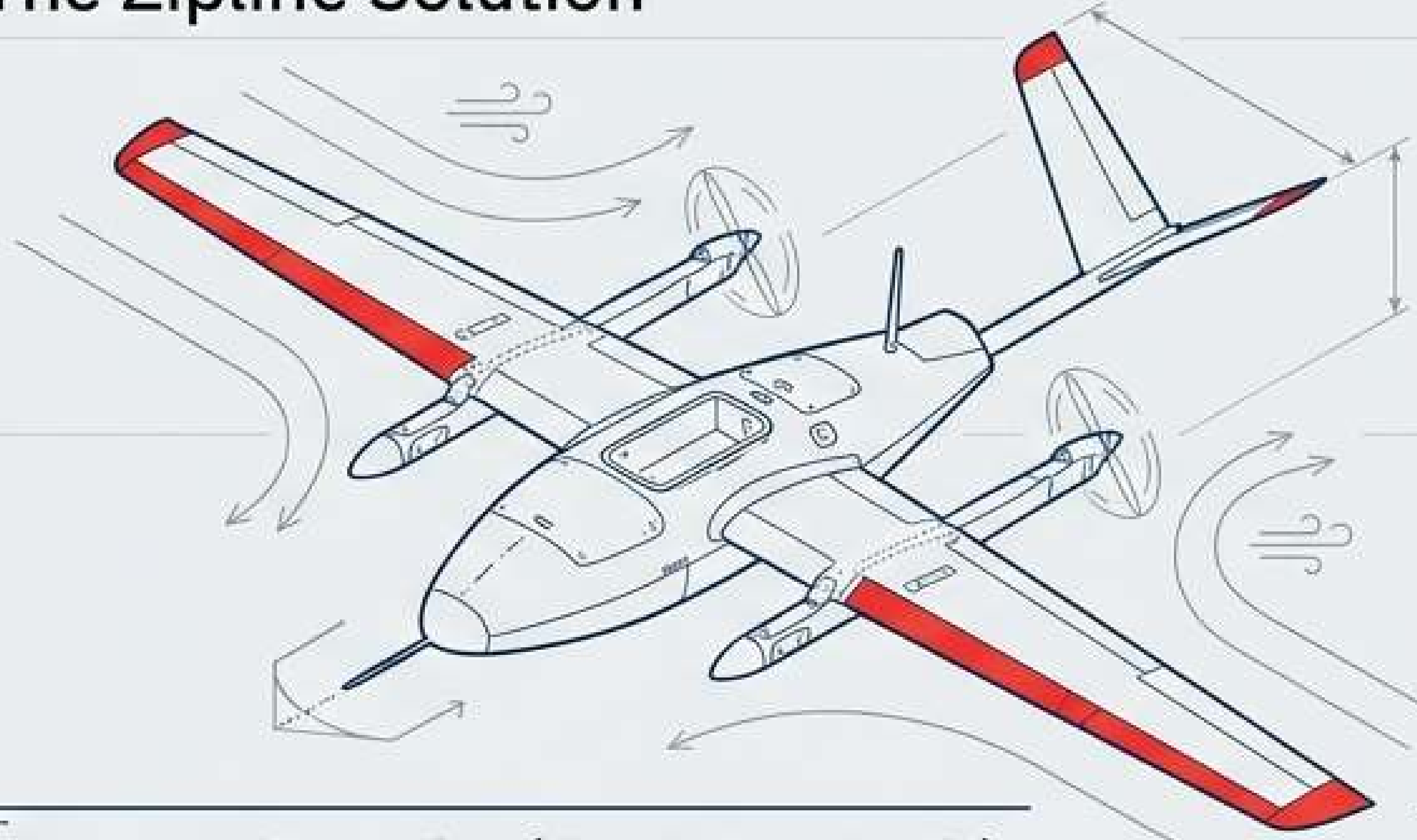


Weight-to-Payload Inefficiency Ratio:

3000:5

Traditional Friction: Vulnerable road networks, high carbon emissions, and highly variable transit times.

The Zipline Solution



Aerospace Bypass: Level 4 autonomous aerial networks replacing ground infrastructure entirely.

The Catalyst

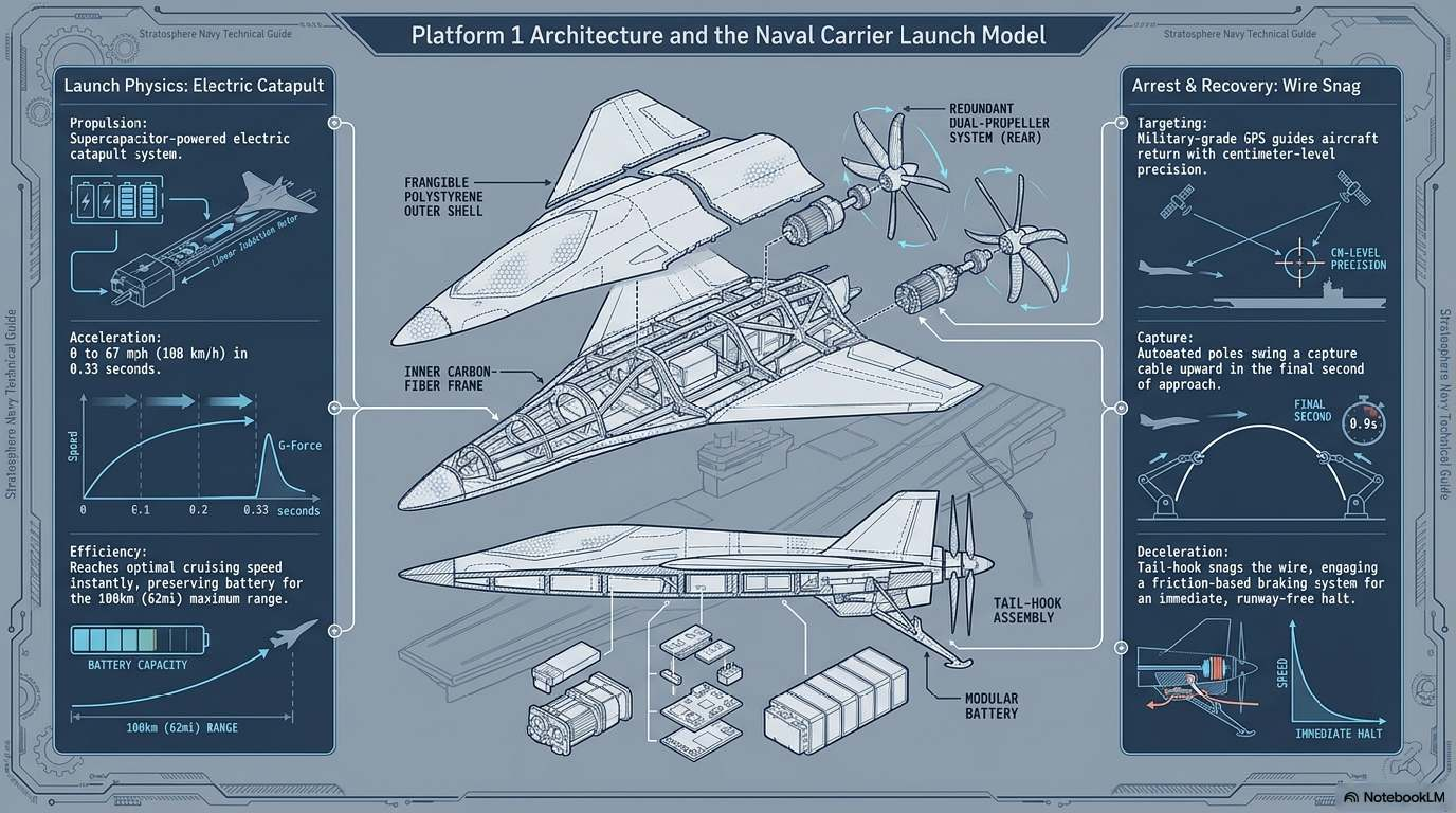
5.5 million children die annually due to lack of access to basic medical products.

The Objective

Build the world's first logistics system serving all people equally, bypassing ground infrastructure.

The Execution

120M+ autonomous flight miles flown, proving the viability of massive-scale instant logistics.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Platform 1 Architecture and the Naval Carrier Launch Model
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Launch Physics: Electric Catapult
Propulsion:
Supercapacitor-powered electric catapult system.
Acceleration:
0 to 67 mph (108 km/h) in 0.33 seconds.
G-Force
0
0.1
0.2
0.33
seconds
Efficiency:
Reaches optimal cruising speed instantly, preserving battery for the 100km (62mi) maximum range.
BATTERY CAPACITY
100km (62mi) RANGE
FRANGIBLE POLYSTYRENE OUTER SHELL
REDUNDANT DUAL-PROPELLER SYSTEM (REAR)
INNER CARBON-FIBER FRANE
TAIL-HOOK ASSEMBLY
MODULAR BATTERY
Arrest & Recovery: Wire Snag
Targeting:
Military-grade GPS guides aircraft return with centimeter-level precision.
CM-LEVEL PRECISION
Capture:
Autoeated poles swing a capture cable upward in the final second of approach.
FINAL SECOND
0.9s
Deceleration:
Tail-hook snags the wire, engaging a friction-based braking system for an immediate, runway-free halt.
SPEED
IMMEDIATE HALT
NotebookLM

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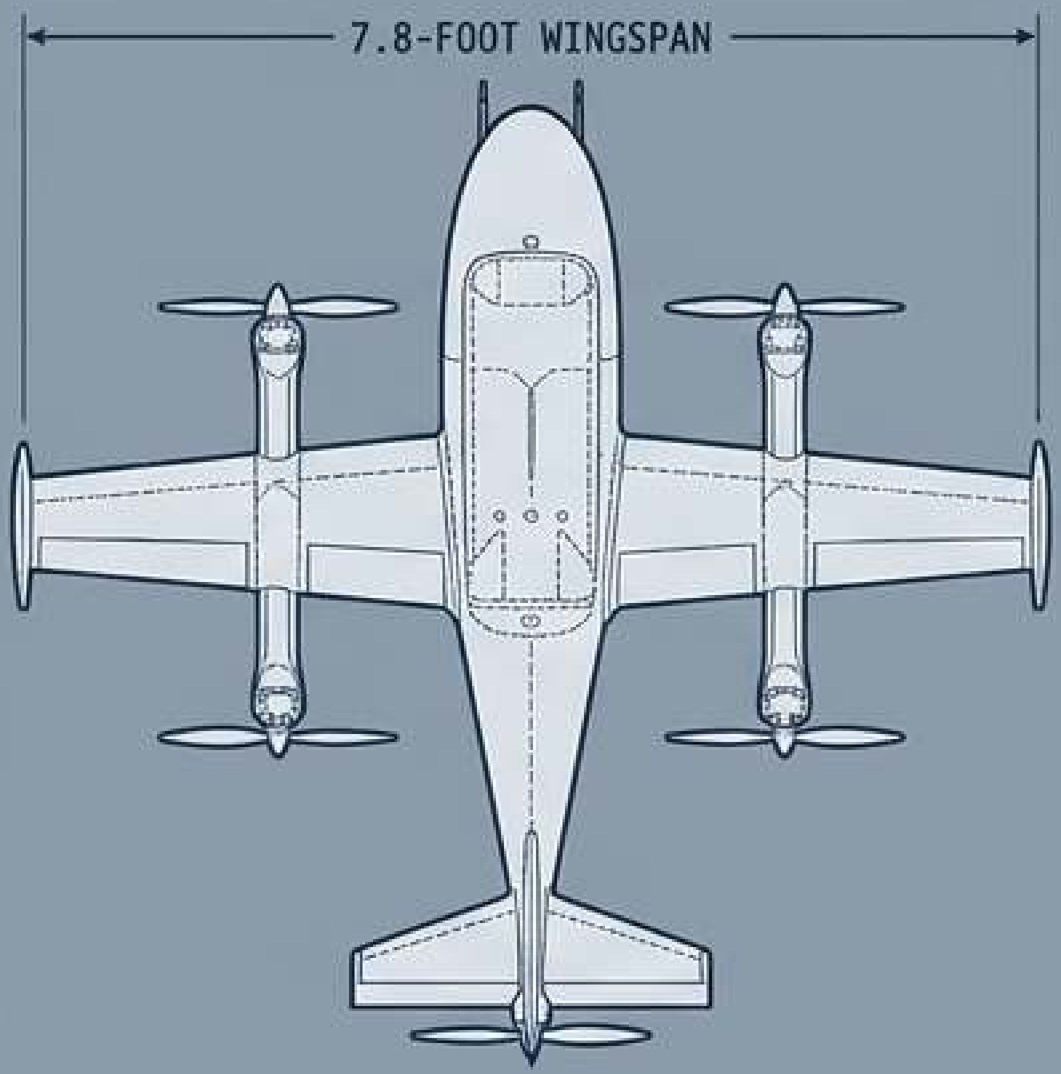
Platform 2: The eVTOL Mothership and Tethered Droid System

TELEMETRY

55-lb Hybrid eVTOL	70 mph Cruise Speed	8-lb Payload Capacity	10-Mile Hub-and-Spoke Radius



Delivery Sequence

01 Acoustic Stealth Hover

Mothership maintains position at 330 ft AGL. Sound signature at ground level is minimized to the volume of rustling leaves.

02 Active Stabilization Descent

The droid is lowered via tether. Three internal fan thrusters actively counter wind shear during descent, offloading positioning work from the mothership.

03 Precision Release

Bottom doors open for dinner-plate (1-meter) accuracy payload release. Total hover time is strictly capped at 75 seconds.

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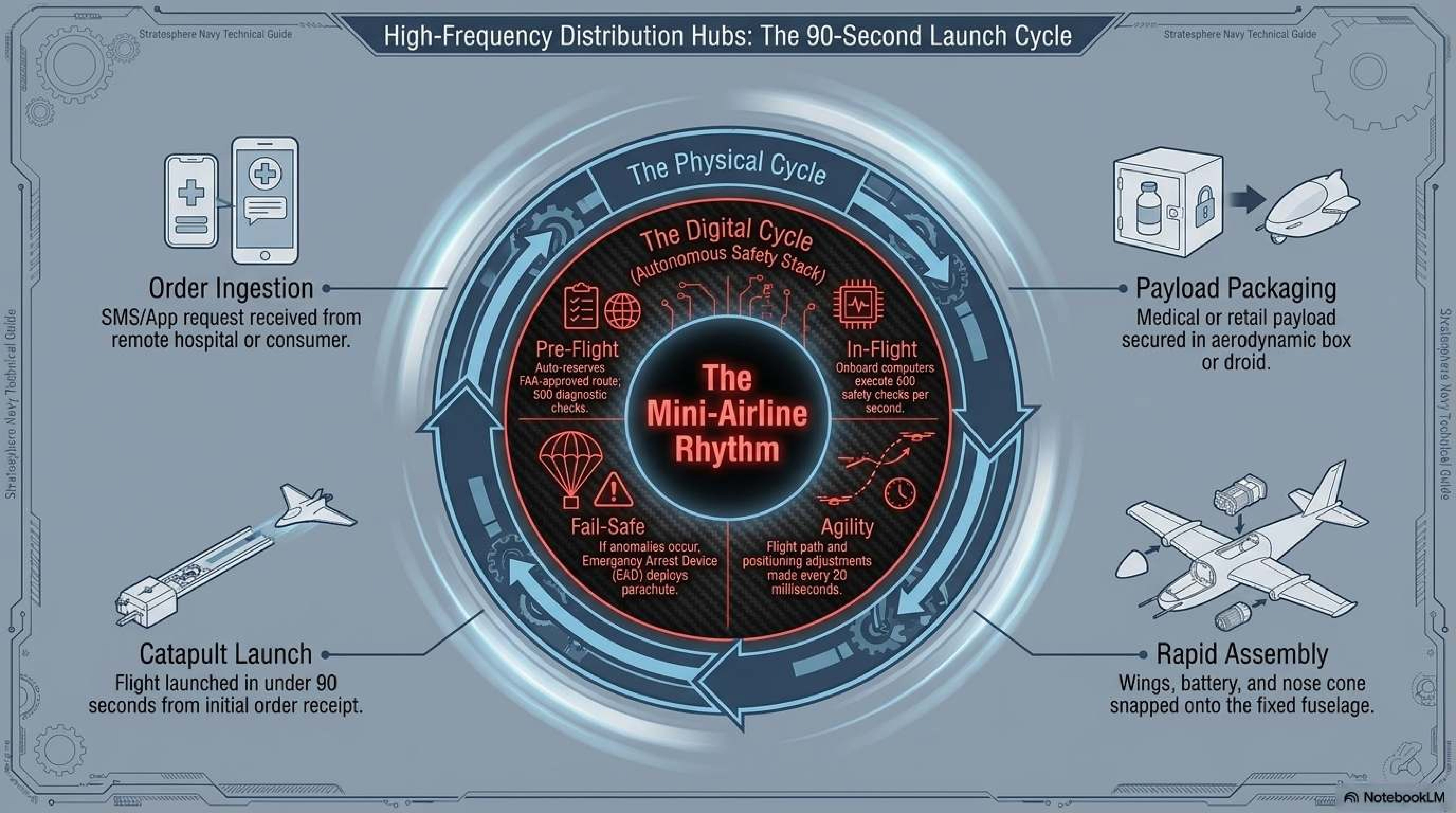

High-Frequency Distribution Hubs: The 90-Second Launch Cycle
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Order Ingestion
SMS/App request received from remote hospital or consumer.
Payload Packaging
Medical or retail payload secured in aerodynamic box or droid.
Rapid Assembly
Wings, battery, and nose cone snapped onto the fixed fuselage.
Catapult Launch
Flight launched in under 90 seconds from initial order receipt.
The Physical Cycle
The Digital Cycle
(Autonomous Safety Stack)
Pre-Flight
Auto-reserves FAA-approved route; 500 diagnostic checks.
In-Flight
Onboard computers execute 600 safety checks per second.
Fail-Safe
If anomalies occur, Emergancy Arrest Device (EAD) deploys parachute.
Agility
Flight path and positioning adjustments made every 20 milliseconds.
The Mini-Airline Rhythm
NotebookLM

African Proving Ground: Optimizing the Rwandan Blood Supply

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System Health

Coverage Optimization

>75%

Of the national blood supply outside Kigali is now delivered by Zipline.

System Health

Velocity Gains

61%

Reduction in blood product delivery times across rural healthcare facilities.

System Health

Waste Reduction

67%

Decrease in blood unit expirations and total medical spoilage.

System Health

Emergency Response

43%

Of all daily flights are direct, rapid responses to critical medical emergencies.

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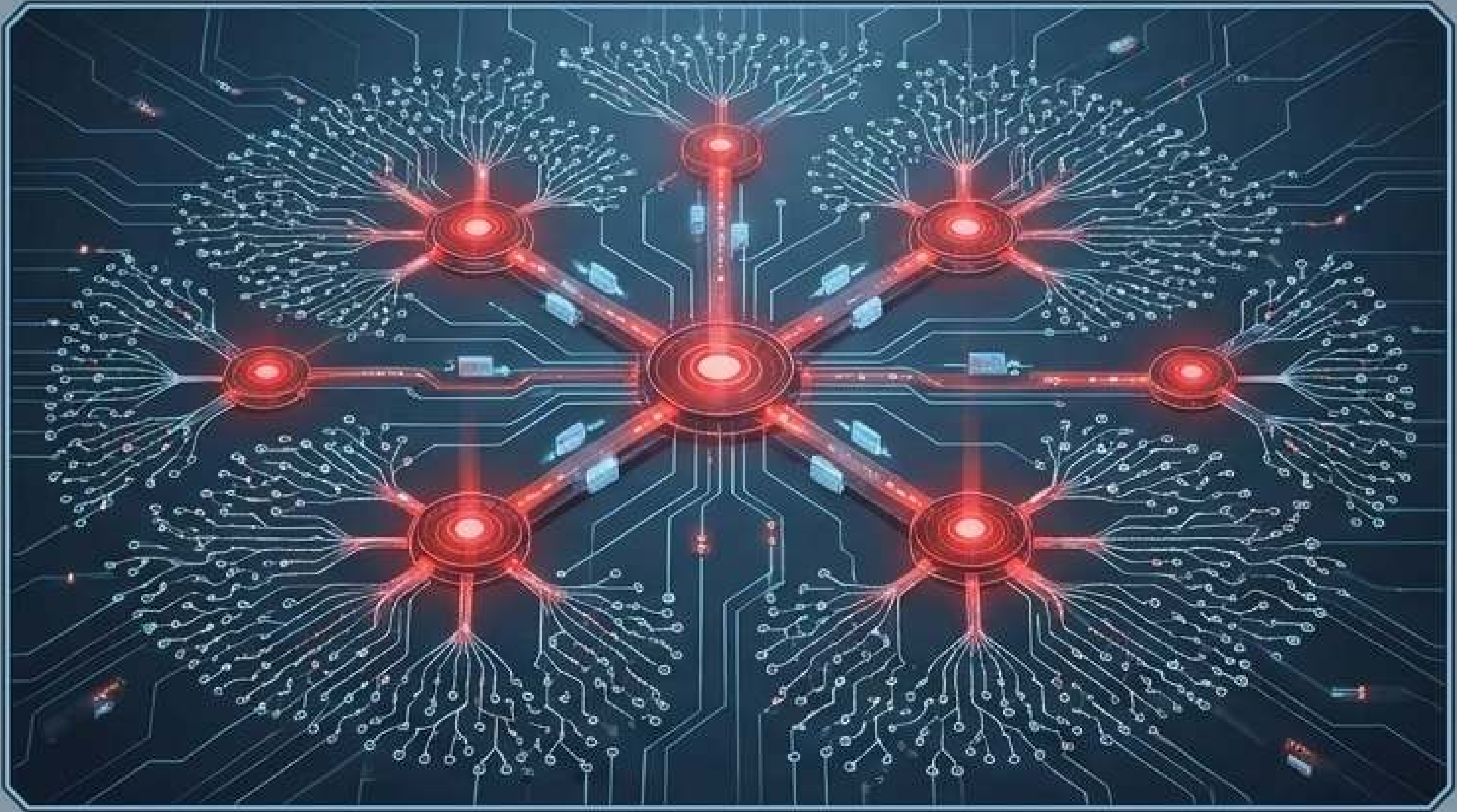
Scaling the Network: Ghana and the Volume Protocol

12 Million+	2,300+	600 Deliveries/Day
Population Covered	Health Facilities Served	Capacity per Hub



Performance Diagnostic

Vaccine Availability
60%

Reduction in facility stock-outs.

Efficiency
42%

Decrease in inventory-driven missed vaccination opportunities.

Resilience
21%

Decrease in total days facilities operated without critical supplies.

Milestone: Delivery of over **1,000,000 COVID-19 vaccine doses** directly to clinics utilizing specialized ultra-low temperature cold-chain payload systems.

The Regulatory Flywheel: From Rwanda to FAA Part 135

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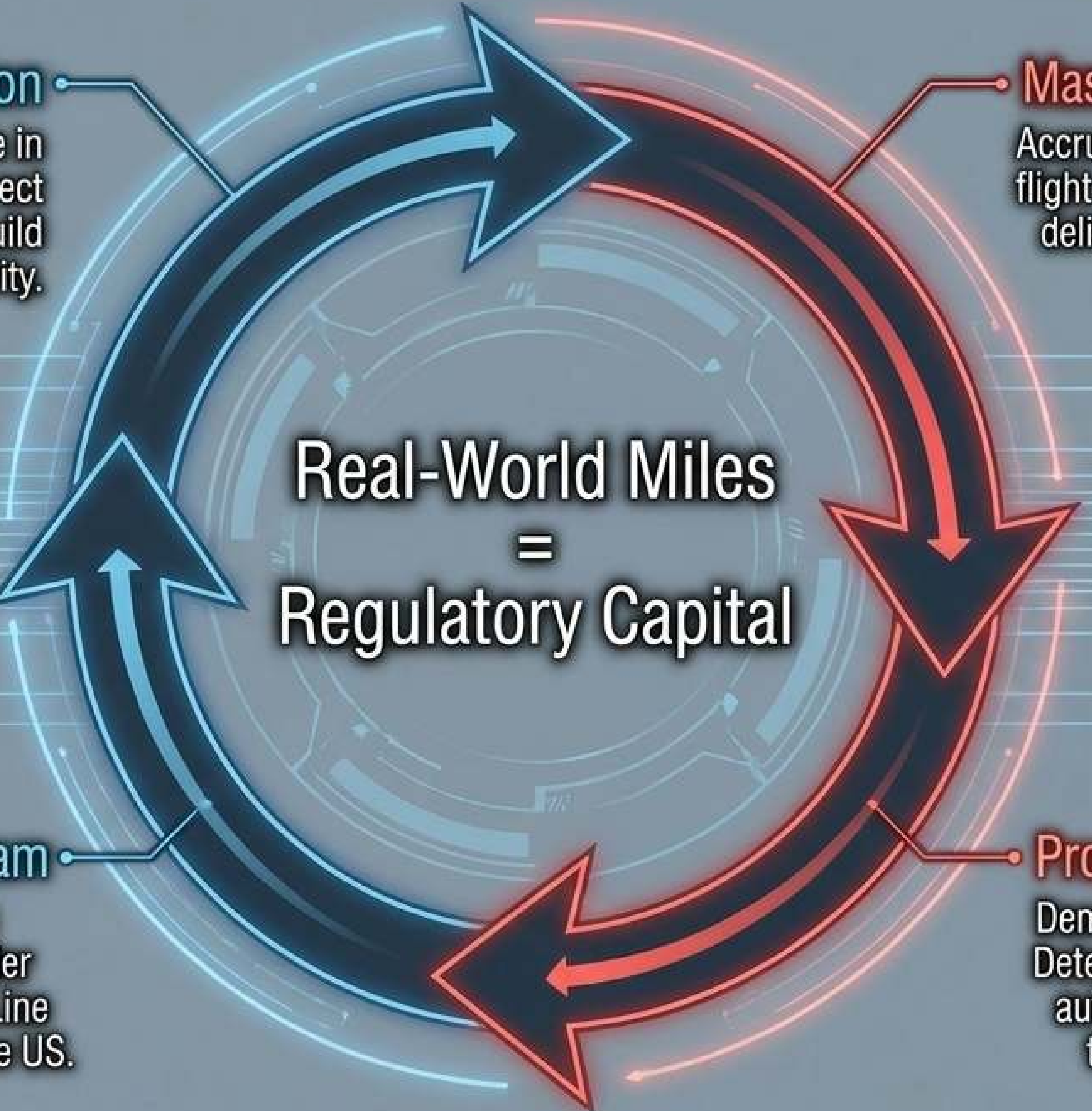
Permissive Incubation

Operate safely at national scale in Rwanda and Ghana, serving direct public health mandates to build operational maturity.

Massive Data Accumulation

Accrue over 120 million autonomous flight miles and 2 million commercial deliveries, creating an unmatched empirical safety dataset.



The FAA BEYOND Program

Leverage safety data to gain rigorous FAA Part 135 Air Carrier Certification for Beyond Visual Line of Sight (BVLOS) operations in the US.

Prove Autonomous Safety

Demonstrate flawless operation of Detect-and-Avoid systems, Level 4 autonomy capabilities, and zero third-party safety incidents.

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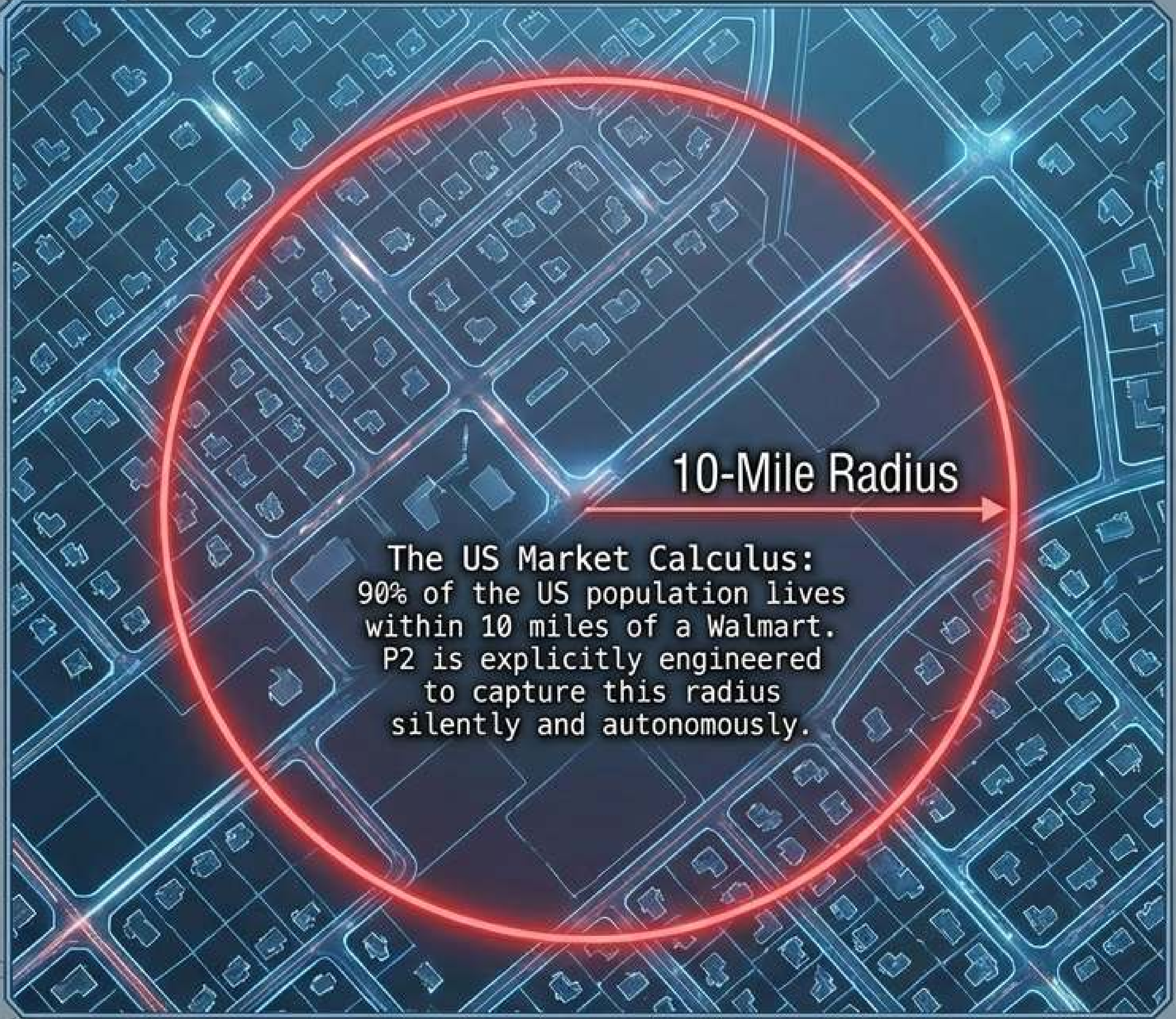
US Commercial Translation: Automating the Suburbs



Commercial Integration

Retail & E-Commerce (Walmart)

Operations scaling in Dallas-Fort Worth and Pea Ridge, Arkansas. Moving everyday retail from 3,000-lb delivery vans to 55-lb precise aerial drops.

Commercial Integration

Healthcare (Intermountain & Novant)

Delivering prescriptions and essential medical supplies directly to residential doorsteps across Utah and North Carolina.

Commercial Integration

Food Services (Instant Delivery)

Expanding rapid instant delivery with partners like sweetgreen, Chipotle, and Panera, proving multi-vertical platform utility.

Global Telemetry: The Autonomous Impact Dashboard

120,000,000+
Autonomous Commercial Miles Flown

2,000,000+
Total Commercial Deliveries

0
Third-Party Incidents

5,000+
Hospitals and Health Centers integrated globally.

5,000,000+
Units of vaccines and medical products delivered.

98%
Reduction in delivery emissions compared to traditional automotive logistics.

Production Velocity: By the end of 2025, Zipline manufacturing scaled to produce one new Zip aircraft every hour to meet network expansion demands.

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The Uncatchable Moat: Lessons in Instant Logistics

Pillar 1: Hardware-Network Symbiosis

Industry:
Trying to build one compromise drone for all potential use cases.

Zipline:
Deploying specialized hardware for specific network densities: P1 (Catapult/Distance for rural) and P2 (eVTOL/Precision for urban).

Pillar 2: The Data Head Start

Industry:
Testing in limited, highly geofenced US suburban pilot zones.

Zipline:
Over a decade of national-scale operations provides an insurmountable lead in edge-case training and machine learning for flight controllers.

Pillar 3: Regulatory Integration

Industry:
Treating regulators as obstacles and reacting to evolving airspace rules.

Zipline:
Operating as an established, certified air carrier (Part 135) with fully integrated FAA traffic management and onboard Detect-and-Avoid.

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Zipline did not just build a delivery drone; they built a Level-4 autonomous mini-airline that replaces ground infrastructure entirely.